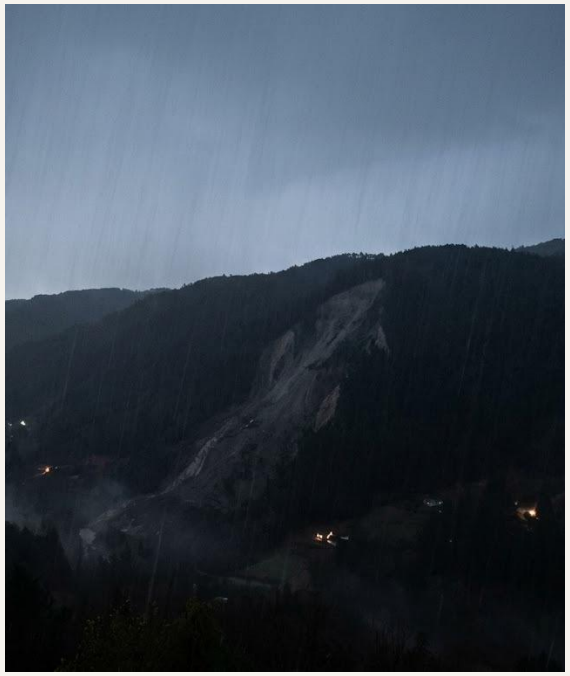
Survivors have hours.

A hillside let go in the dark. The Army needed 31 hours to get in.



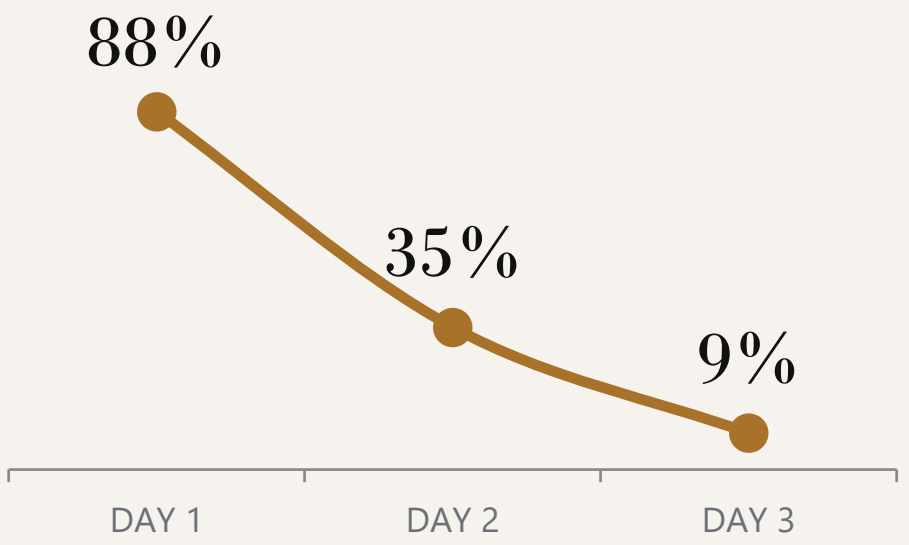
02:17 · 30 JULY 2024
WAYANAD, KERALA

1.23 M
lives lost to disasters, 2000–2019

355
lives lost in Kerala's rain disasters, 2024

Chance a trapped person is still alive

REPORTED · 1990 PHILIPPINES EARTHQUAKE



Reported figures.

Buildings can be rebuilt. People can't.

Too far. Too slow. Too silent.

"…leaving trapped survivors isolated and search-and-rescue teams operating blind." PS S.3



TOO FAR

Blind from orbit

India's space chief: deep underground can't be detected from space.

ISRO chairman S. Somanath · Aug 2024



TOO SLOW

~1,500 on foot

Wayanad was searched by hand. Nobody was found.

Wayanad · July–Aug 2024



TOO SILENT

95.2% offline

of Puerto Rico's cell sites after Hurricane Maria.

FCC · Hurricane Maria, 2017

Between space and the ground, the sky is empty.

RESQMESH

When the towers fall, the sky becomes the network.

PS S.3 · DECENTRALIZED DRONE SWARM FOR SEARCH & RESCUE

EYES

2 Scouts · fly low

ANSWERS · TOO FAR

Spot every survivor.

Find trapped people. Pin their GPS location.

BRAIN

2 Mappers · mid-height

ANSWERS · TOO SLOW

Plan the search together.

Split the zone so nothing is searched twice.

VOICE

1 Relay-Anchor · high

ANSWERS · TOO SILENT

Carry every pin home.

A flying mesh. No tower. No internet.



HOW ONE PIN TRAVELS

Person seen → GPS pin → Drone mesh → Relay-Anchor → Long-range link → Rescue base → Rescue teams

The swarm is the network.

	DRONE SEARCH TODAY	RESQMESH
FLIES	One drone, flown by hand.	→ An autonomous swarm.
CONNECTS	Needs towers or a pilot link.	→ The swarm is the network.
DELIVERS	Raw video to watch.	→ GPS survivor pins.

To our knowledge, no operational system combines all three.

How it works.

01 → **Sense**
SCOUTS
Camera detects people.
GPS stamps each find.

02 → **Plan**
MAPPERS
Split the zone.
Plan energy-aware paths.

03 → **Link**
RELAY-ANCHOR
Pins hop the mesh,
even if a link drops.

04 **Show**
RESCUE BASE
Offline dashboard.
Rescuers follow GPS.

TECH STACK

ROS 2
swarm software

Unreal Engine
disaster simulation

Dashboard
live web map and pins

GPS
survivor coordinates

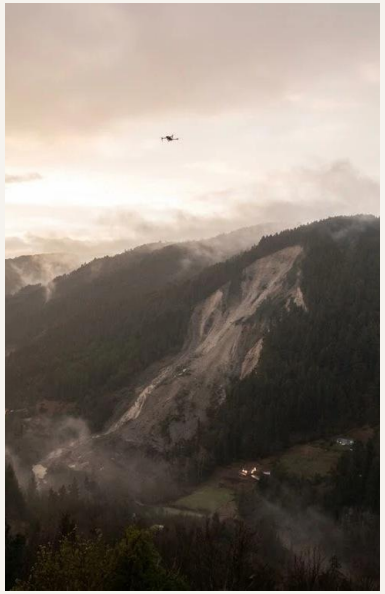
Fusion
custom drone design

Lose a drone, and the swarm re-plans.

Proof, not a promise.

A live demo in a simulated disaster, built in two 16-hour sprints.

- **SPRINT I** 16 HOURS
  Unreal world · 5 ROS 2 drones
  planner · dashboard
- **SPRINT II** 16 HOURS
  Detect → pin → relay
  survive a drone loss
- **AFTER THE HACKATHON**
  Custom drones → field trial



WHY IT IS FEASIBLE

- ✓ Simulation first, no hardware risk
- ✓ Proven open tools: ROS 2, Unreal
- ✓ Five drones, scoped to the brief

THE LIVE DEMO

Disaster → survivors found →
pins on the dashboard →
drone lost → swarm re-plans.

At 02:17, nothing was in the air.

Next time, something will be.